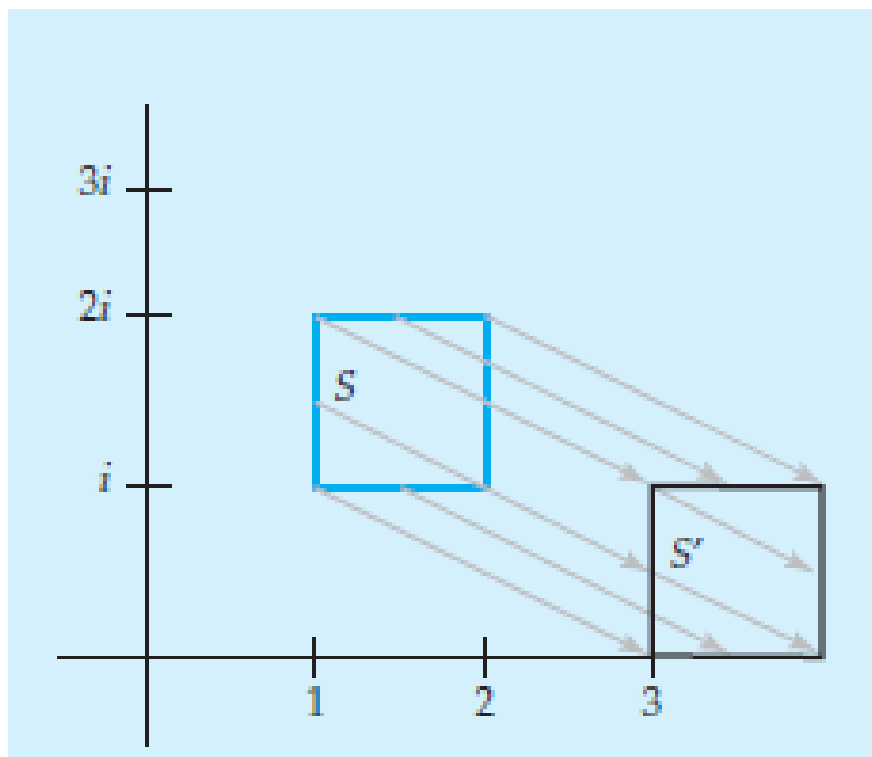




COMPLEX FUNCTION

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Complex Function

A complex function is a function f whose domain and range are subsets of the set C of complex numbers.

EXAMPLE:

The expression $z^2 - (2 + i)z$ can be evaluated at any complex number z and always yields a single complex number, and so $f(z) = z^2 - (2 + i)z$ defines a complex function. Values of f are found by using the arithmetic operations for complex numbers given in Section 1.1. For instance, at the points $z = i$ and $z = 1 + i$

we have: $f(i) = (i)^2 - (2 + i)(i) = -1 - 2i + 1 = -2i$

And $f(1 + i) = (1 + i)^2 - (2 + i)(1 + i) = 2i - 1 - 3i = -1 - i$.

Complex Function

A symbol such as z , which can stand for any one of a set of complex number is called as **Complex Variable**.

EXAMPLE:

The expression $z^2 - (2 + i)z$ can be evaluated at any complex number z and always yields a single complex number.

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ELEMENTARY FUNCTIONS

1. **Polynomial Functions** are defined by

$$w = a_0 z^n + a_1 z^{n-1} + \cdots + a_{n-1} z + \boxed{a_n} = P(z) \quad (2.2)$$

where $a_0 \neq 0$, $a_1, \dots, a_n$ are complex constants and n is a positive integer called the *degree* of the polynomial $P(z)$.

The transformation $w = az + b$ is called a linear transformation.

2. **Rational Algebraic Functions** are defined by

$$\boxed{w = \frac{P(z)}{Q(z)}} \quad (2.3)$$

where $P(z)$ and $Q(z)$ are polynomials. We sometimes call (2.3) a *rational transformation*. The special case $w = (az + b)/(cz + d)$ where $ad - bc \neq 0$ is often called a bilinear or fractional linear transformation.

3. **Exponential Functions** are defined by

$$\underline{w = e^z = e^{x+iy} = e^x(\cos y + i \sin y)} \quad (2.4)$$

where e is the *natural base of logarithms*. If a is real and positive, we define

$$\boxed{a^z = e^{z \ln a}} \quad (2.5)$$

where $\ln a$ is the *natural logarithm of a* . This reduces to (4) if $a = e$.

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ELEMENTARY FUNCTIONS

Complex exponential functions have properties similar to those of real exponential functions. For example, $e^{z_1} \cdot e^{z_2} = e^{z_1+z_2}$, $e^{z_1}/e^{z_2} = e^{z_1-z_2}$.

4. Trigonometric Functions. We define the trigonometric or circular functions $\sin z$, $\cos z$, etc., in terms of exponential functions as follows:

$$\left[\begin{array}{ll} \sin z = \frac{e^{iz} - e^{-iz}}{2i}, & \cos z = \frac{e^{iz} + e^{-iz}}{2} \\ \sec z = \frac{1}{\cos z} = \frac{2}{e^{iz} + e^{-iz}}, & \csc z = \frac{1}{\sin z} = \frac{2i}{e^{iz} - e^{-iz}} \\ \tan z = \frac{\sin z}{\cos z} = \frac{e^{iz} - e^{-iz}}{i(e^{iz} + e^{-iz})}, & \cot z = \frac{\cos z}{\sin z} = \frac{i(e^{iz} + e^{-iz})}{e^{iz} - e^{-iz}} \end{array} \right.$$

Many of the properties familiar in the case of real trigonometric functions also hold for the complex trigonometric functions. For example, we have:

$$\left[\begin{array}{l} \sin^2 z + \cos^2 z = 1, \quad 1 + \tan^2 z = \sec^2 z, \quad 1 + \cot^2 z = \csc^2 z \\ \sin(-z) = -\sin z, \quad \cos(-z) = \cos z, \quad \tan(-z) = -\tan z \\ \sin(z_1 \pm z_2) = \sin z_1 \cos z_2 \pm \cos z_1 \sin z_2 \\ \cos(z_1 \pm z_2) = \cos z_1 \cos z_2 \mp \sin z_1 \sin z_2 \\ \tan(z_1 \pm z_2) = \frac{\tan z_1 \pm \tan z_2}{1 \mp \tan z_1 \tan z_2} \end{array} \right.$$

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ELEMENTARY FUNCTIONS

5. **Hyperbolic Functions** are defined as follows:

$$\begin{aligned} \sinh z &= \frac{e^z - e^{-z}}{2}, & \cosh z &= \frac{e^z + e^{-z}}{2} \\ \operatorname{sech} z &= \frac{1}{\cosh z} = \frac{2}{e^z + e^{-z}}, & \operatorname{csch} z &= \frac{1}{\sinh z} = \frac{2}{e^z - e^{-z}} \\ \tanh z &= \frac{\sinh z}{\cosh z} = \frac{e^z - e^{-z}}{e^z + e^{-z}}, & \coth z &= \frac{\cosh z}{\sinh z} = \frac{e^z + e^{-z}}{e^z - e^{-z}} \end{aligned}$$

The following properties hold:

$$\begin{aligned} \cosh^2 z - \sinh^2 z &= 1, & 1 - \tanh^2 z &= \operatorname{sech}^2 z, & \coth^2 z - 1 &= \operatorname{csch}^2 z \\ \sinh(-z) &= -\sinh z, & \cosh(-z) &= \cosh z, & \tanh(-z) &= -\tanh z \\ \sinh(z_1 \pm z_2) &= \sinh z_1 \cosh z_2 \pm \cosh z_1 \sinh z_2 \\ \cosh(z_1 \pm z_2) &= \cosh z_1 \cosh z_2 \pm \sinh z_1 \sinh z_2 \\ \tanh(z_1 \pm z_2) &= \frac{\tanh z_1 \pm \tanh z_2}{1 \pm \tanh z_1 \tanh z_2} \end{aligned}$$

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ELEMENTARY FUNCTIONS

The following relations exist between the trigonometric or circular functions and the hyperbolic functions:

$$\begin{array}{lll} \sin iz = i \sinh z, & \cos iz = \cosh z, & \tan iz = i \tanh z \\ \sinh iz = i \sin z, & \cosh iz = \cos z, & \tanh iz = i \tan z \end{array}$$

6. **Logarithmic Functions.** If $z = e^w$, then we write $w = \ln z$, called the *natural logarithm* of z . Thus the natural logarithmic function is the inverse of the exponential function and can be defined by

$$w = \ln z = \ln r + i(\theta + 2k\pi), \quad k = 0, \pm 1, \pm 2, \dots$$

where $z = re^{i\theta} = re^{i(\theta+2k\pi)}$. Note that $\ln z$ is a multiple-valued (in this case, infinitely-many-valued) function. The *principal-value* or *principal branch* of $\ln z$ is sometimes defined as $\ln r + i\theta$ where $0 \leq \theta < 2\pi$. However, any other interval of length 2π can be used, e.g., $-\pi < \theta \leq \pi$, etc.

The logarithmic function can be defined for real bases other than e . Thus, if $z = a^w$, then $w = \log_a z$ where $a > 0$ and $a \neq 1$. In this case, $z = e^{w \ln a}$ and so, $w = (\ln z)/(\ln a)$.

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ELEMENTARY FUNCTIONS

7. **Inverse Trigonometric Functions.** If $z = \sin w$, then $w = \sin^{-1} z$ is called the *inverse sine* of z or *arc sine* of z . Similarly, we define other inverse trigonometric or circular functions $\cos^{-1} z$, $\tan^{-1} z$, etc. These functions, which are multiple-valued, can be expressed in terms of natural logarithms as follows. In all cases, we omit an additive constant $2k\pi i$, $k = 0, \pm 1, \pm 2, \dots$, in the logarithm:

$$\sin^{-1} z = \frac{1}{i} \ln \left(iz + \sqrt{1 - z^2} \right),$$

$$\csc^{-1} z = \frac{1}{i} \ln \left(\frac{i + \sqrt{z^2 - 1}}{z} \right)$$

$$\cos^{-1} z = \frac{1}{i} \ln \left(z + \sqrt{z^2 - 1} \right),$$

$$\sec^{-1} z = \frac{1}{i} \ln \left(\frac{1 + \sqrt{1 - z^2}}{z} \right)$$

$$\tan^{-1} z = \frac{1}{2i} \ln \left(\frac{1 + iz}{1 - iz} \right),$$

$$\cot^{-1} z = \frac{1}{2i} \ln \left(\frac{z + i}{z - i} \right)$$

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ELEMENTARY FUNCTIONS

8. **Inverse Hyperbolic Functions.** If $z = \sinh w$, then $w = \sinh^{-1} z$ is called the *inverse hyperbolic sine* of z . Similarly, we define other inverse hyperbolic functions $\cosh^{-1} z$, $\tanh^{-1} z$, etc. These functions, which are multiple-valued, can be expressed in terms of natural logarithms as follows. In all cases, we omit an additive constant $2k\pi i$, $k = 0, \pm 1, \pm 2, \dots$, in the logarithm:

$$\left[\begin{array}{ll} \sinh^{-1} z = \ln(z + \sqrt{z^2 + 1}), & \operatorname{csch}^{-1} z = \ln\left(\frac{1 + \sqrt{z^2 + 1}}{z}\right) \\ \cosh^{-1} z = \ln(z + \sqrt{z^2 - 1}), & \operatorname{sech}^{-1} z = \ln\left(\frac{1 + \sqrt{1 - z^2}}{z}\right) \\ \tanh^{-1} z = \frac{1}{2} \ln\left(\frac{1 + z}{1 - z}\right), & \operatorname{coth}^{-1} z = \frac{1}{2} \ln\left(\frac{z + 1}{z - 1}\right) \end{array} \right.$$

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BASIC DEFINITIONS

Distance: $|z - z_0|$ represents the distance between two points z and z_0 .

Circle: $|z - z_0| = r$ represents a circle with centre at the point z_0 and radius r .

Interior of a circle: $|z - z_0| < r$ represents the interior of the circle.

Exterior of a circle: $|z - z_0| > r$ represents the exterior of the circle.

Annulus: The region between two concentric circles of radii r_1 and r_2 ($r_2 > r_1$) and centre at z_0 is known as the annulus region and is represented as

$$r_1 < |z - z_0| < r_2$$

Neighbourhood: The set of all points for which $|z - z_0| < r$ is known as the neighbourhood of z_0 .

Boundary point: A point which does not lie in the interior or exterior of a region is known as a boundary point.

Open set: A set that does not contain its boundary points is known as an open set.

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BASIC DEFINITIONS

Closed set: A set that contains all its boundary points is known as a closed set.

Connected set: If any two points of the set can be joined by a polygonal line such that all the points of the line also belong to the set then the set is known as a connected set.

Domain: A set which is open and connected is known as a domain.

Bounded region: A region which can be enclosed in a circle of finite radius is known as a bounded region.

Compact region: A region that is closed and bounded is known as a compact region.

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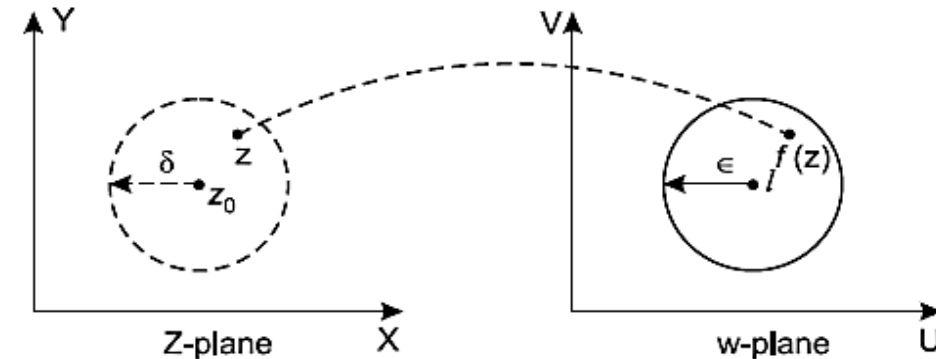
LIMIT OF FUNCTION OF COMPLEX VARIABLE

Let $f(z)$ be a single valued function defined at all points in some neighbourhood of point z_0 . Then $f(z)$ is said to have the limit l as z approaches z_0 along any path if given an arbitrary real number $\epsilon > 0$, however small there exists a real number $\delta > 0$, such that

$$|f(z) - l| < \epsilon \text{ whenever } 0 < |z - z_0| < \delta$$

i.e. for every $z \neq z_0$ in δ -disc (dotted) of z -plane, $f(z)$ has a value lying in the ϵ -disc of w -plane

In symbolic form, $\lim_{z \rightarrow z_0} f(z) = l$



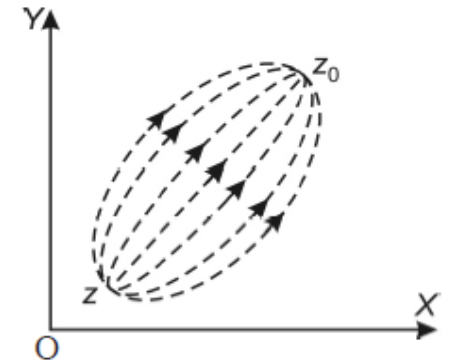
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LIMIT OF FUNCTION OF COMPLEX VARIABLE

Note: (I) δ usually depends upon ϵ .

(II) $z \rightarrow z_0$ implies that z approaches z_0 along any path. The limits must be independent of the manner in which z approaches z_0 .

If we get two different limits as $z \rightarrow z_0$ along two different paths then limits does not exist.



Example:

(1) Find Following Limit, $\lim_{z \rightarrow i} \frac{z^2 + 1}{z^6 + 1}$

As $z \rightarrow i, z^2 \rightarrow -1$

$$\begin{aligned} \therefore \lim_{z \rightarrow i} \frac{z^2 + 1}{z^6 + 1} &= \lim_{z \rightarrow i} \frac{z^2 + 1}{(z^2 + 1)(z^4 - z^2 + 1)} \\ &= \lim_{z^2 \rightarrow -1} \frac{1}{(z^4 - z^2 + 1)} = \frac{1}{3} \end{aligned}$$

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LIMIT OF FUNCTION OF COMPLEX VARIABLE

~~Example (2):~~ Show that $\lim_{z \rightarrow 0} \frac{z}{|z|}$ does not exist.

Solution. $\lim_{z \rightarrow 0} \frac{z}{|z|} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x + iy}{\sqrt{x^2 + y^2}}$

Let $y = mx$,

$$= \lim_{x \rightarrow 0} \frac{x + imx}{\sqrt{x^2 + m^2 x^2}} = \lim_{x \rightarrow 0} \frac{1 + im}{\sqrt{1 + m^2}} = \frac{1 + mi}{\sqrt{1 + m^2}}$$

The value of $\frac{1 + mi}{\sqrt{1 + m^2}}$ are different for different values of m .

Hence, limit of the function does not exist.

An Epsilon-Delta Proof of a Limit

EXAMPLE 3: Prove that $\lim_{z \rightarrow 1+i} (2+i)z = 1+3i$.

Solution According to Definition 2.8, $\lim_{z \rightarrow 1+i} (2+i)z = 1+3i$, if, for every $\varepsilon > 0$, there is a $\delta > 0$ such that $|(2+i)z - (1+3i)| < \varepsilon$ whenever $0 < |z - (1+i)| < \delta$. Proving that the limit exists requires that we find an appropriate value of δ for a given value of ε . In other words, for a given value of ε we must find a positive number δ with the property that if $0 < |z - (1+i)| < \delta$, then $|(2+i)z - (1+3i)| < \varepsilon$. One way of finding δ is to “work backwards.” The idea is to start with the inequality:

$$|(2+i)z - (1+3i)| < \varepsilon \quad (4)$$

★
z - limit
< delta

lhs - rhs <
epsilon

An Epsilon-Delta Proof of a Limit

EXAMPLE 3: Prove that $\lim_{z \rightarrow 1+i} (2+i)z = 1+3i$.

and then use properties of complex numbers and the modulus to manipulate this inequality until it involves the expression $|z - (1+i)|$. Thus, a natural first step is to factor $(2+i)$ out of the left-hand side of (4):

$$|2+i| \cdot \left| z - \frac{1+3i}{2+i} \right| < \varepsilon. \quad (5)$$

Because $|2+i| = \sqrt{5}$ and $\frac{1+3i}{2+i} = 1+i$, (5) is equivalent to:

$$\sqrt{5} \cdot |z - (1+i)| < \varepsilon \quad \text{or} \quad |z - (1+i)| < \frac{\varepsilon}{\sqrt{5}}. \quad (6)$$

An Epsilon-Delta Proof of a Limit

EXAMPLE 3: Prove that $\lim_{z \rightarrow 1+i} (2+i)z = 1+3i$.

Thus, (6) indicates that we should take $\delta = \varepsilon/\sqrt{5}$. Keep in mind that the choice of δ is not unique. Our choice of $\delta = \varepsilon/\sqrt{5}$ is a result of the particular algebraic manipulations that we employed to obtain (6). Having found δ we now present the formal proof that $\lim_{z \rightarrow 1+i} (2+i)z = 1+3i$ that does not indicate how the choice of δ was made:

[Given $\varepsilon > 0$, let $\delta = \varepsilon/\sqrt{5}$. If $0 < |z - (1+i)| < \delta$, then we have $|z - (1+i)| < \varepsilon/\sqrt{5}$.] Multiplying both sides of the last inequality by $|2+i| = \sqrt{5}$ we obtain:

$$|2+i| \cdot |z - (1+i)| < \sqrt{5} \cdot \frac{\varepsilon}{\sqrt{5}} \quad \text{or} \quad |(2+i)z - (1+3i)| < \varepsilon.$$

Therefore, $|(2+i)z - (1+3i)| < \varepsilon$ whenever $0 < |z - (1+i)| < \delta$. So, according to Definition 2.8, we have proven that $\lim_{z \rightarrow 1+i} (2+i)z = 1+3i$.

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CONTINUITY

The function $f(z)$ of a complex variable z is said to be continuous at the point z_0 if for any given positive number ϵ , we can find a number δ such that $|f(z) - f(z_0)| < \epsilon$ for all points z of the domain satisfying

$$|z - z_0| < \delta$$

$f(z)$ is said to be continuous at $z = z_0$ if
$$\lim_{z \rightarrow z_0} f(z) = f(z_0)$$

CONTINUITY IN TERMS OF REAL & IMAGINARY NUMBER :-

If $w = f(z) = u(x, y) + iv(x, y)$ is continuous function at $z = z_0$ then $u(x, y)$ and $v(x, y)$ are separately continuous functions of x, y at (x_0, y_0) where $z_0 = x_0 + iy_0$.

Conversely, if $u(x, y)$ and $v(x, y)$ are continuous functions of x, y at (x_0, y_0) then $f(z)$ is continuous at $z = z_0$.

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CONTINUITY

Example 1

Show that the function $f(z)$ defined by

$$f(z) = \begin{cases} \frac{\operatorname{Re}(z)}{z} & , z \neq 0 \\ 0 & , z = 0 \end{cases} \quad \text{is not continuous at } z = 0$$

Solution. Here $f(z) = \frac{\operatorname{Re}(z)}{z}$ when $z \neq 0$

$$\lim_{z \rightarrow 0} \frac{\operatorname{Re}(z)}{z} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x}{x + iy} = \lim_{x \rightarrow 0} \left[\lim_{y \rightarrow 0} \frac{x}{x + iy} \right] = \lim_{x \rightarrow 0} \frac{x}{x} = 1$$

Again

$$\lim_{z \rightarrow 0} \frac{\operatorname{Re}(z)}{z} = \lim_{y \rightarrow 0} \left[\lim_{x \rightarrow 0} \frac{x}{x + iy} \right] = 0$$

As $z \rightarrow 0$, for two different paths limit have two different values. So, limit does not exist.
Hence $f(z)$ is not continuous at $z = 0$ **Proved.**

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CONTINUITY

Example 2

Discuss the continuity of $f(z)$ at the origin.

$$f(z) = \frac{\bar{z}}{z}, \quad z \neq 0$$
$$= 0, \quad z = 0$$

Solution

$$\lim_{z \rightarrow 0} f(z) = \lim_{z \rightarrow 0} \frac{\bar{z}}{z}$$
$$= \lim_{z \rightarrow 0} \frac{x - iy}{x + iy}$$

Let $z \rightarrow 0$ along the line $y = mx$.

$$\lim_{z \rightarrow 0} f(z) = \lim_{x \rightarrow 0} \frac{x - imx}{x + imx}$$
$$= \lim_{x \rightarrow 0} \frac{1 - im}{1 + im}$$
$$= \frac{1 - im}{1 + im}$$

Since the limit depends on m , it takes different values along different paths. Thus, the limit does not exist. Hence, $f(z)$ is not continuous at the origin.

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CONTINUITY

Theorems on Continuity

- THEOREM 2.2.** Given $f(z)$ and $g(z)$ are continuous at $z = z_0$. Then so are the functions $f(z) + g(z)$, $f(z) - g(z)$, $f(z)g(z)$ and $f(z)/g(z)$, the last if $g(z_0) \neq 0$. Similar results hold for continuity in a region.
- THEOREM 2.3.** Among the functions continuous in every finite region are (a) all polynomials, (b) e^z , (c) $\sin z$ and $\cos z$.
- THEOREM 2.4.** Suppose $w = f(z)$ is continuous at $z = z_0$ and $z = g(\zeta)$ is continuous at $\zeta = \zeta_0$. If $z_0 = g(\zeta_0)$, then the function $w = f[g(\zeta)]$, called a *function of a function* or *composite function*, is continuous at $\zeta = \zeta_0$. This is sometimes briefly stated as: A continuous function of a continuous function is continuous.
- THEOREM 2.5.** Suppose $f(z)$ is continuous in a closed and bounded region. Then it is bounded in the region; i.e., there exists a constant M such that $|f(z)| < M$ for all points z of the region.
- THEOREM 2.6.** If $f(z)$ is continuous in a region, then the real and imaginary parts of $f(z)$ are also continuous in the region.

Problems on Limit and Continuity

- EXAMPLE 4:**
- (a) Prove that $f(z) = z^2$ is continuous at $z = z_0$.
- (b) Prove that $f(z) = \begin{cases} z^2 & z \neq z_0 \\ 0 & z = z_0 \end{cases}$, where $z_0 \neq 0$, is discontinuous at $z = z_0$.

Solution

(a) By Problem 2.23(a), $\lim_{z \rightarrow z_0} f(z) = f(z_0) = z_0^2$ and so $f(z)$ is continuous at $z = z_0$.

Another Method. We must show that given any $\epsilon > 0$, we can find $\delta > 0$ (depending on ϵ) such that $|f(z) - f(z_0)| = |z^2 - z_0^2| < \epsilon$ when $|z - z_0| < \delta$. The proof patterns that given in Problem 2.23(a).

(b) By Problem 2.23(b), $\lim_{z \rightarrow z_0} f(z) = z_0^2$, but $f(z_0) = 0$. Hence, $\lim_{z \rightarrow z_0} f(z) \neq f(z_0)$ and so $f(z)$ is discontinuous at $z = z_0$ if $z_0 \neq 0$.

★ If $z_0 = 0$, then $f(z) = 0$; and since $\lim_{z \rightarrow z_0} f(z) = 0 = f(0)$, we see that the function is continuous.

Problems on Limit and Continuity

 **EXAMPLE 5:** Is the function $f(z) = \frac{3z^4 - 2z^3 + 8z^2 - 2z + 5}{z - i}$ continuous at $z = i$?

Solution

$f(i)$ does not exist, i.e., $f(x)$ is not defined at $z = i$. Thus $f(z)$ is not continuous at $z = i$.

By redefining $f(z)$ so that $f(i) = \lim_{z \rightarrow i} f(z) = 4 + 4i$ (see Problem 2.25), it becomes continuous at $z = i$. In such a case, we call $z = i$ a removable discontinuity.

EXAMPLE 6:

Prove that $f(z) = z^2$ is continuous in the region $|z| \leq 1$.

Solution

Let z_0 be any point in the region $|z| \leq 1$. By Problem 2.23(a), $f(z)$ is continuous at z_0 . Thus, $f(z)$ is continuous in the region since it is continuous at any point of the region.

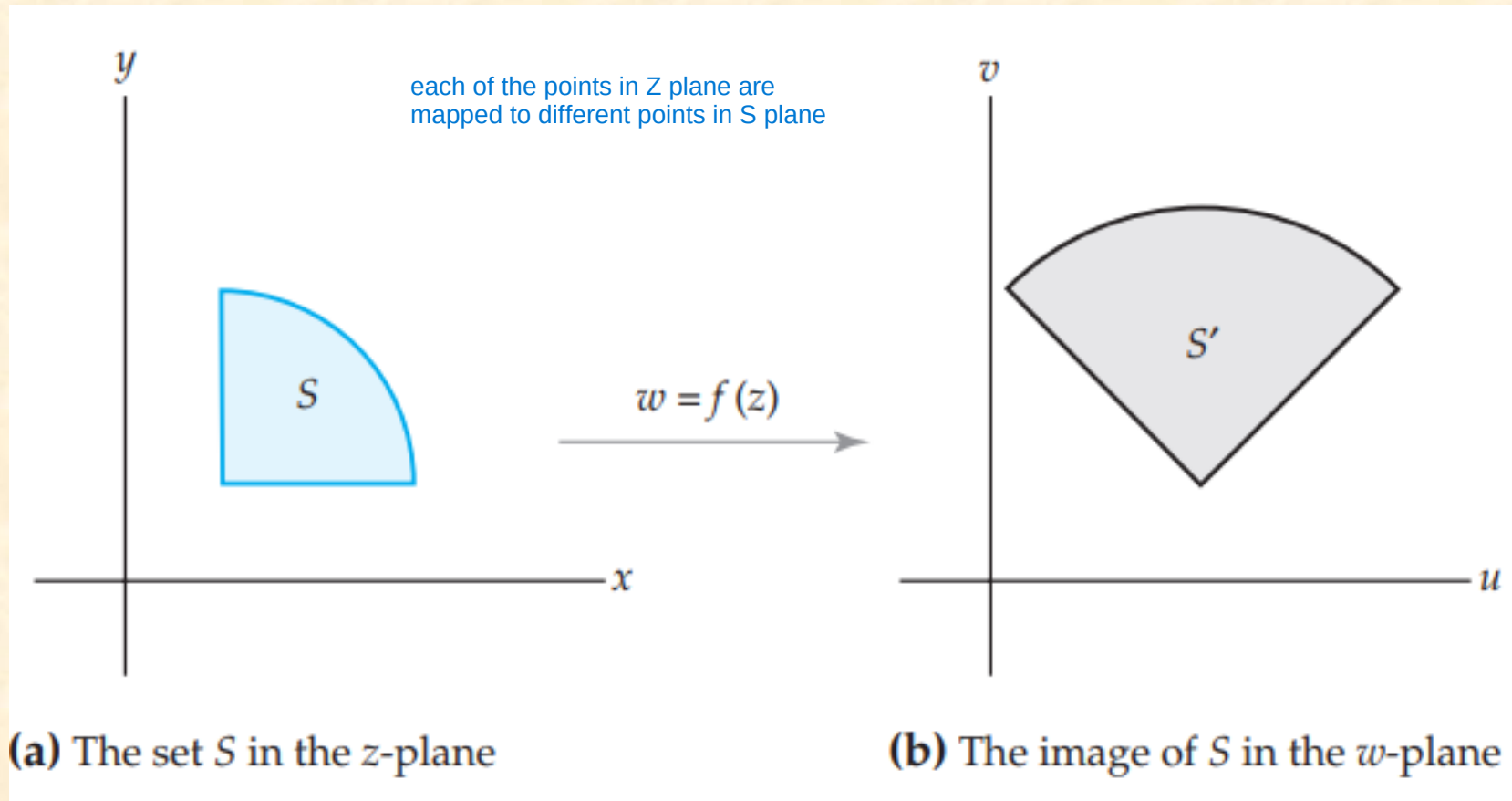
Complex Functions as Mappings

Mappings A useful tool for the study of real functions in elementary calculus is the graph of the function. Recall that if $y = f(x)$ is a real-valued function of a real variable x , then the graph of f is defined to be the set of all points $(x, f(x))$ in the two-dimensional Cartesian plane. An analogous definition can be made for a complex function. However, if $w = f(z)$ is a complex function, then both z and w lie in a complex plane. It follows that the set of all points $(z, f(z))$ lies in four-dimensional space (two dimensions from the input z and two dimensions from the output w). Of course, a subset of four-dimensional space cannot be easily illustrated.

Complex Functions as Mappings

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Complex Functions as Mappings



Complex Functions as Mappings

EXAMPLE 7: Let $w = f(z) = z^2$. Find the values of w that correspond to (a) $z = -2 + i$ and (b) $z = 1 - 3i$, and show how the correspondence can be represented graphically.

Solution

(a) $w = f(-2 + i) = (-2 + i)^2 = 4 - 4i + i^2 = 3 - 4i$

(b) $w = f(1 - 3i) = (1 - 3i)^2 = 1 - 6i + 9i^2 = -8 - 6i$

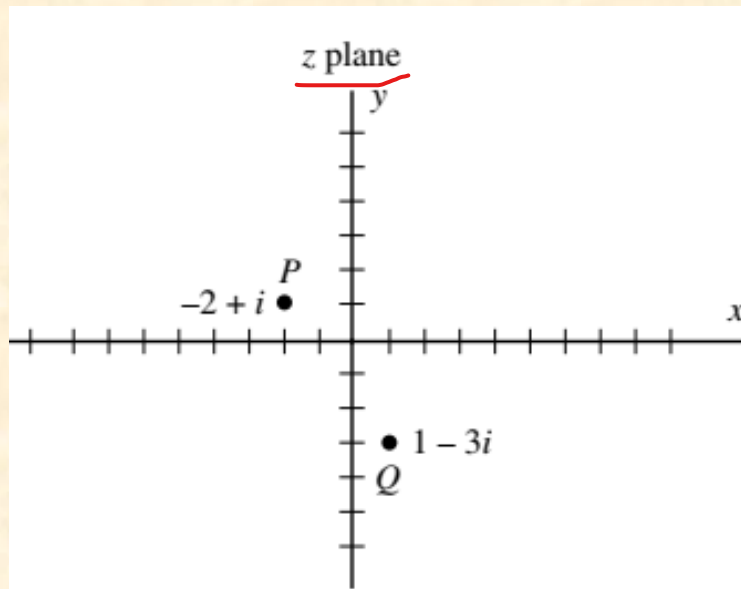


Fig. 2-6

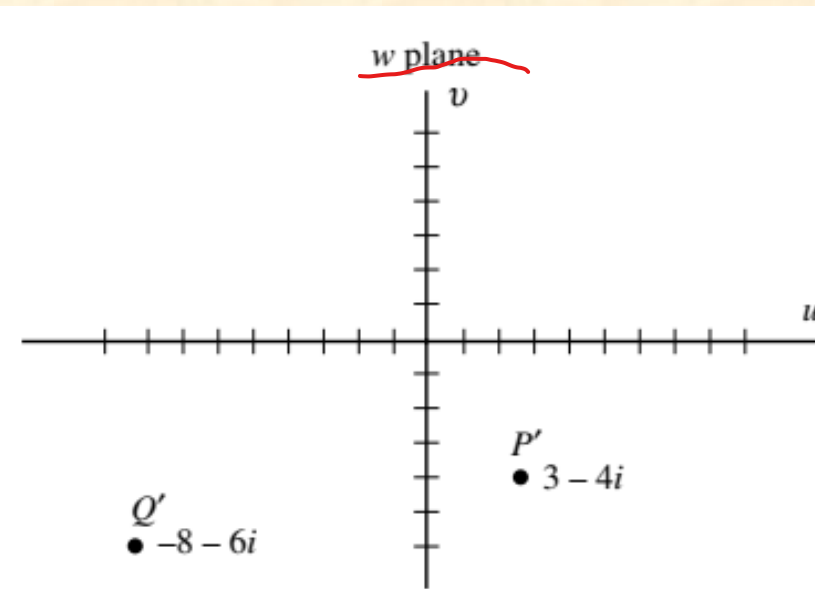


Fig. 2-7

Complex Functions as Mappings

EXAMPLE 8: A point P moves in a counterclockwise direction around a circle in the z plane having center at the origin and radius 1. If the mapping function is $w = z^3$, show that when P makes one complete revolution, the image P' of P in the w plane makes three complete revolutions in a counterclockwise direction on a circle having center at the origin and radius 1.

Solution

Let $z = re^{i\theta}$. Then, on the circle $|z| = 1$ [Fig. 2-8], $r = 1$ and $z = e^{i\theta}$. Hence, $w = z^3 = (e^{i\theta})^3 = e^{3i\theta}$. Letting (ρ, ϕ) denote polar coordinates in the w plane, we have $w = \rho e^{i\phi} = e^{3i\theta}$ so that $\rho = 1$, $\phi = 3\theta$.

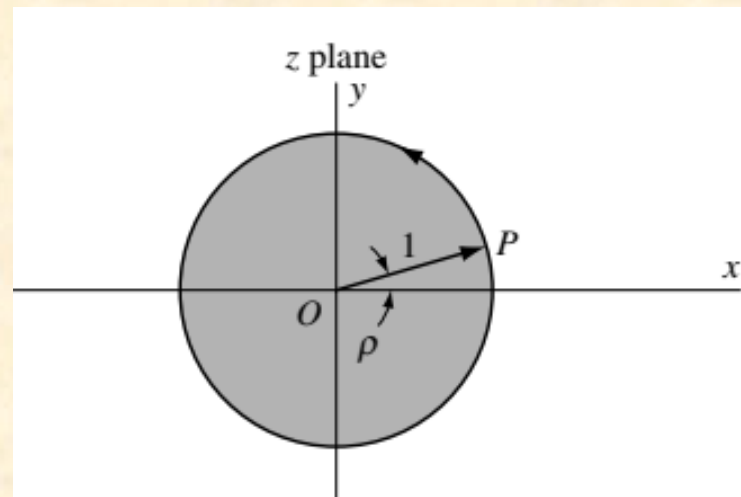


Fig. 2-8

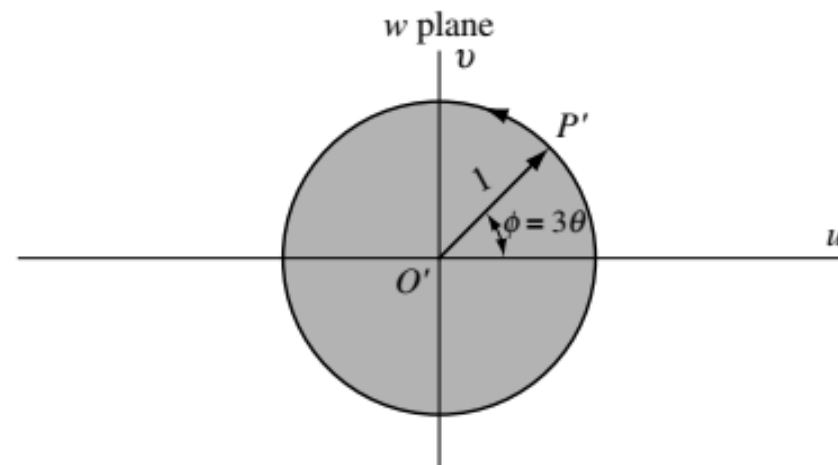


Fig. 2-9

Complex Functions as Mappings

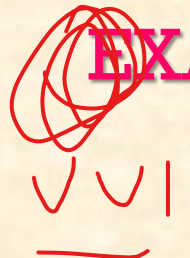
EXAMPLE 8: A point P moves in a counterclockwise direction around a circle in the z plane having center at the origin and radius 1. If the mapping function is $w = z^3$, show that when P makes one complete revolution, the image P' of P in the w plane makes three complete revolutions in a counterclockwise direction on a circle having center at the origin and radius 1.

Solution

Let $z = re^{i\theta}$. Then, on the circle $|z| = 1$ [Fig. 2-8], $r = 1$ and $z = e^{i\theta}$. Hence, $w = z^3 = (e^{i\theta})^3 = e^{3i\theta}$. Letting (ρ, ϕ) denote polar coordinates in the w plane, we have $w = \rho e^{i\phi} = e^{3i\theta}$ so that $\rho = 1$, $\phi = 3\theta$.

Since $\rho = 1$, it follows that the image point P' moves on a circle in the w plane of radius 1 and center at the origin [Fig. 2-9]. Also, when P moves counterclockwise through an angle θ , P' moves counterclockwise through an angle 3θ . Thus, when P makes one complete revolution, P' makes three complete revolutions. In terms of vectors, it means that vector $O'P'$ is rotating three times as fast as vector OP .

Complex Functions as Mappings

 **EXAMPLE 9:** Find the image of the vertical line $x = 1$ under the complex mapping $w = z^2$ and represent the mapping graphically.

Solution Let C be the set of points on the vertical line $x = 1$ or, equivalently, the set of points $z = 1 + iy$ with $-\infty < y < \infty$. since $z = x + iy$ We proceed as in Example 1. From (1) of Section 2.1, the real and imaginary parts of $w = z^2$ are $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$, respectively. For a point $z = 1 + iy$ in C , we have $u(1, y) = 1 - y^2$ and $v(1, y) = 2y$. This implies that the image of S is the set of points $w = u + iv$ satisfying the simultaneous equations:

$$u = 1 - y^2 \quad \text{real part} \quad (3)$$

$$v = 2y \quad \text{imaginary} \quad (4)$$

and

Complex Functions as Mappings

we map from parametric equations to cartesian equations because they offer a direct relationship between coordinates.

EXAMPLE 9: Find the image of the vertical line $x = 1$ under the complex mapping $w = z^2$ and represent the mapping graphically.

for $-\infty < y < \infty$. Equations (3) and (4) are *parametric equations* in the *real parameter y* , and they define a curve in the w -plane. We can find a *Cartesian equation in u and v* for this curve by eliminating the parameter y . In order to do so, we solve (4) for y and then substitute this expression into (3):

create a cartesian equation from

u and v (real and imaginary parts/axis)

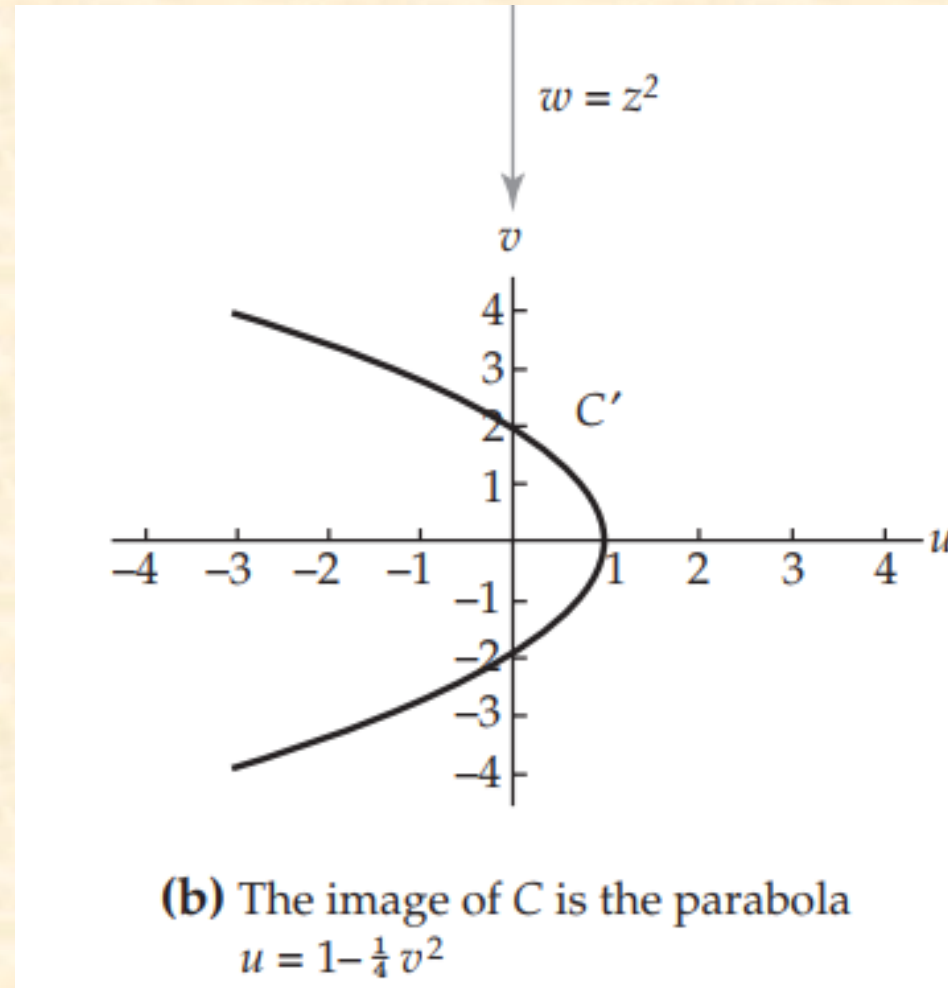
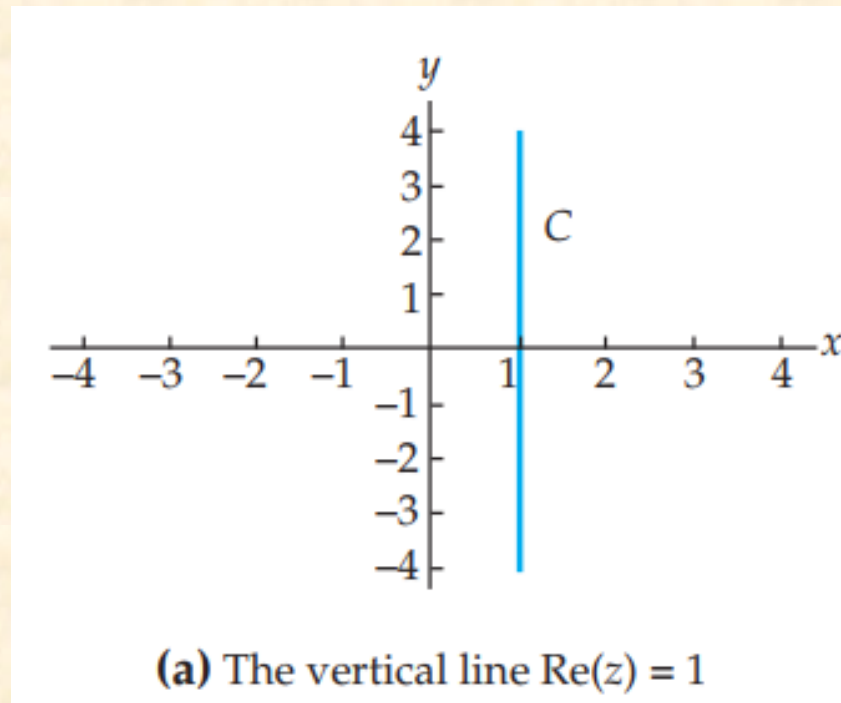
to map on the complex plane

$$u = 1 - \left(\frac{v}{2}\right)^2 = 1 - \frac{v^2}{4}. \quad (5)$$

Since y can take on any real value and since $v = 2y$, it follows that v can take on any real value in (5). Consequently, C' —the image of C —is a parabola in the w -plane with vertex at $(1,0)$ and u -intercepts at $(0, \pm 2)$. See Figure 2.3(b). In conclusion, we have shown that the vertical line $x = 1$ shown in color in Figure 2.3(a) is mapped onto the parabola $u = 1 - \frac{1}{4}v^2$ shown in black in Figure 2.3(b) by the complex mapping $w = z^2$.

Complex Functions as Mappings

EXAMPLE 9: Find the image of the vertical line $x = 1$ under the complex mapping $w = z^2$ and represent the mapping graphically.



Thanks a lot ...